

Experiment 1: Case Study

Problem Statement: Choose KL ERP System, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World Case Study: KL ERP System

1. Functional Requirements

Student

- Register/Login
- Course Registration
- View Attendance
- View Timetable
- Pay Fees
- View Results
- Apply Leave

Faculty

- Manage Attendance
- Upload Marks
- Manage Courses

Admin

- Manage Users
- Manage Departments
- Generate Reports

2. User Stories

User ID	User Story	Priority
US-01	As a student, I want to login securely.	High
US-02	As a student, I want to register courses.	High
US-03	As a student, I want to view attendance.	High
US-04	As a student, I want to pay fees online.	High

US-05	As a student, I want to view results.	Medium
US-06	As a faculty member, I want to upload marks.	High
US-07	As an admin, I want to manage users.	High
US-08	As an admin, I want to generate reports.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reasons:

- Frequent requirement changes
- Continuous feedback
- Faster delivery
- Easy maintenance

4. Programming Language Selection

Backend: Java (Spring Boot)

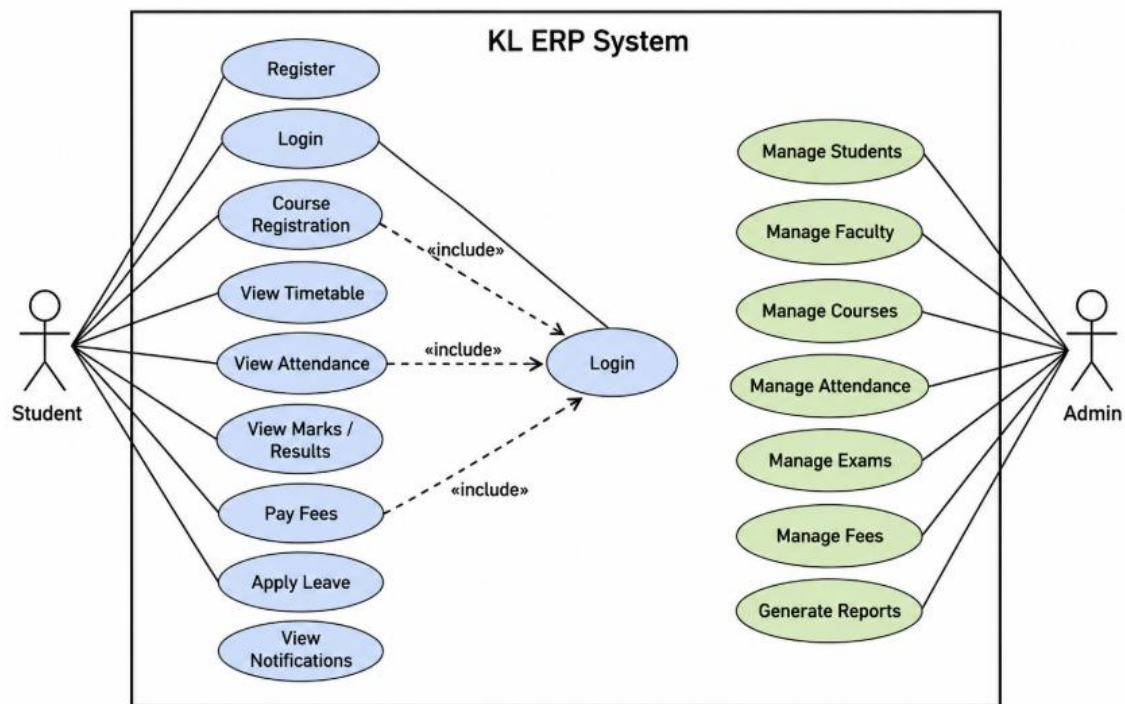
Frontend: HTML5, CSS3, JavaScript, React.js

Database: MySQL

Other: REST API, JWT, Git, Docker, AWS

5. System Architecture

Student/Faculty/Admin → Web Application → Spring Boot REST API → MySQL Database



6. Implementation Strategy

Requirement Analysis
System Design
Development
Testing
Deployment

7. Evaluation Strategy

Functional Accuracy
Performance
Security
Reliability
Usability
Scalability
Maintainability

8. Expected Outcomes

- Efficient ERP management
- Secure student information
- Faster academic processes
- Improved user satisfaction
- Scalable architecture